Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster for Small Satellites

Team: Repulsor Mark - 1
Sidhwa Tanuj Haresh - 230003073 - Mechanical Engineering
Tejas Santosh Bhavekar - 230003079 - Mechanical Engineering
Siddharth Vezzu - 230021016 - Space Science and Engineering
Sudarsanam Sai Sri Narasimha - 230021018 - Space Science and Engineering

Abstract

Accurate prediction of compressible flow phenomena is critical for the design and optimization of micro cold-gas thrusters in small satellite applications. This report details the development, validation, and application of a quasi-one-dimensional numerical solver for the Euler equations to analyze flow through converging-diverging nozzles. Three finite-volume schemes; Lax-Friedrichs, HLLC, and a second-order MUSCL-HLLC formulation;were implemented and evaluated across subsonic, shock-containing, and fully supersonic expansion regimes. Rigorous verification and validation protocols, including grid independence studies, exact analytical comparisons, and 2D axisymmetric ANSYS Fluent simulations, confirmed the MUSCL-HLLC scheme (employing a MinMod flux limiter) as the most accurate method for resolving sharp shock discontinuities with minimal numerical diffusion. The validated solver was subsequently utilized for a parametric performance evaluation of a scaled 1 mm^2 micro-thruster. The effects of nozzle geometry and propellant species) were quantified. The results successfully captured the classical propulsion trade-off: Helium achieved a superior specific impulse of $\sim$ 244 s, while Nitrogen produced a higher absolute thrust of $\sim$ 495 mN. The developed computational framework demonstrates high-fidelity shock-capturing capabilities at a fraction of the computational cost of multi-dimensional CFD, providing a highly reliable tool for preliminary space propulsion design.

Keywords: Compressible flow, Nozzle flow, HLLC, MUSCL, Shock waves, Cold-gas thruster, Finite volume method

Problem Statement

The rapid miniaturization of spacecraft has led to the widespread adoption of small satellites and CubeSats, which rely heavily on micro cold-gas thrusters for precise attitude control and orbital maneuvering. Due to strict volume and mass constraints, these micro-propulsion systems operate at extremely small physical scales—often with throat areas on the order of a single square millimeter. Designing and optimizing these nozzles requires an accurate understanding of compressible flow physics, including transonic acceleration, shock wave formation in over-expanded regimes, and supersonic expansion in the vacuum of space.

While commercial Computational Fluid Dynamics (CFD) software can resolve these complex multi-dimensional flows, utilizing them for preliminary parametric design iterations is often computationally prohibitive and time-consuming. Conversely, relying solely on exact analytical isentropic equations fails to capture shock discontinuities and entropy-generating phenomena accurately under varying back-pressure conditions.

Therefore, the primary objective of this project is to bridge this gap by developing a robust, computationally efficient quasi-one-dimensional numerical solver from scratch. This custom in-house physics engine is specifically designed to solve the Euler equations for compressible flow, enabling rapid performance analysis of varying converging-diverging nozzle contours and propellant species. By implementing advanced shock-capturing finite volume schemes—specifically a second-order MUSCL-HLLC formulation with Total Variation Diminishing (TVD) flux limiters—this work aims to create a highly accurate mathematical testbed. Ultimately, this numerical model serves as a foundational tool to predict key thruster performance metrics, including mass flow rate, absolute thrust, and specific impulse, thereby streamlining the micro-propulsion design process.

1 Introduction

The design and analysis of converging-diverging nozzles play a crucial role in the performance of propulsion systems, particularly in micro cold-gas thrusters used for small satellites and CubeSats. Accurate prediction of compressible flow behavior, including shock formation and expansion phenomena, is essential for optimizing thrust and efficiency.

In this work, a numerical solver is developed to simulate quasi-one-dimensional compressible flow through a converging-diverging nozzle by solving the Euler equations in conservative form. The study focuses on capturing key flow features such as transonic acceleration, normal shock formation, and supersonic expansion under varying operating conditions. [1, 2]

To achieve this, three finite-volume based numerical schemes are implemented and systematically compared: the first-order Lax-Friedrichs scheme, the HLLC (Harten-Lax-van Leer Contact) approximate Riemann solver, and a higher-order MUSCL-HLLC scheme with flux limiting for improved spatial accuracy. These methods are evaluated in terms of stability, numerical dissipation, and their ability to resolve discontinuities.

A parametric study is conducted to investigate the influence of back pressure and nozzle geometry on flow behavior and shock location. Two nozzle configurations are considered: a piecewise linear (double-cone) nozzle and a smooth quadratic nozzle.

The numerical results are validated against analytical solutions derived from isentropic flow relations and normal shock theory, as well as against results obtained from ANSYS Fluent simulations. This combined analysis provides a comprehensive assessment of the accuracy and robustness of the implemented numerical methods.

2 Mathematical Formulation & Numerical Methodology

The flow through the converging-diverging nozzle is modeled as a quasi-one-dimensional, compressible, inviscid flow with varying cross-sectional area. Under these assumptions, the governing equations are the Euler equations expressed in conservative form, coupled with numerical methods to handle spatial discretization, flux calculation, and time integration.

2.1 Governing Equations

The conservation equations for mass, momentum, and energy are given by [3]:

$$\frac{\partial(\rho A)}{\partial t} + \frac{\partial(\rho u A)}{\partial x} = 0 \tag{1}$$

$$\frac{\partial(\rho u A)}{\partial t} + \frac{\partial\left((\rho u^2 + p)A\right)}{\partial x} = p\frac{dA}{dx} \tag{2}$$

$$\frac{\partial(EA)}{\partial t} + \frac{\partial\left(u(E+p)A\right)}{\partial x} = 0 \tag{3}$$

where ρ is density, u is velocity, p is pressure, $A(x)$ is the cross-sectional area, and E is the total energy per unit volume. The source term $p\frac{dA}{dx}$ arises due to the variation in nozzle area and represents the effect of geometric changes on the momentum equation.

The system is closed using the ideal gas equation of state:

$$p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2\right) \tag{4}$$

where γ is the ratio of specific heats. The speed of sound and Mach number are defined as:

$$a = \sqrt{\frac{\gamma p}{\rho}}, \qquad M = \frac{u}{a} \tag{5}$$

For numerical implementation, the governing equations are written in a compact vector form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S} \tag{6}$$

where

$$\mathbf{U} = \begin{bmatrix} \rho A \\ \rho u A \\ EA \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u A \\ (\rho u^2 + p)A \\ u(E+p)A \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ p\frac{dA}{dx} \\ 0 \end{bmatrix} \tag{7}$$

2.2 Finite Volume Discretization

The quasi-one-dimensional Euler equations are discretized spatially using a cell-centered finite volume formulation. [4] Integrating the compact vector form over a computational cell i yields the semi-discrete ordinary differential equation:

$$\frac{d\mathbf{U}_i}{dt} = -\frac{\mathbf{F}_{i+\frac{1}{2}} - \mathbf{F}_{i-\frac{1}{2}}}{\Delta x} + \mathbf{S}_i \tag{8}$$

where $\mathbf{U}_i$ represents the cell-averaged conserved variables, $\mathbf{F}_{i\pm 1/2}$ denotes the numerical fluxes at the cell interfaces, and $\mathbf{S}_i$ is the geometric source term evaluated at the cell center.

2.3 Numerical Flux Schemes

To evaluate the interface fluxes, two different schemes are considered to balance robustness and shock-capturing fidelity.

Lax–Friedrichs Scheme: Used as a baseline method due to its simplicity, the interface flux is computed as:

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{F}_L + \mathbf{F}_R\right) - \frac{\alpha}{2}\left(\mathbf{U}_R - \mathbf{U}_L\right) \tag{9}$$

where $\alpha = \max(|u| + a)$ is the maximum local wave speed. While highly robust, this scheme introduces significant numerical dissipation, leading to smeared shock profiles.

HLLC Approximate Riemann Solver: To improve shock resolution and eliminate excessive numerical diffusion, the HLLC scheme is employed. Unlike Lax–Friedrichs, HLLC resolves the three-wave structure of the solution, including the contact discontinuity. [5] The interface flux is given by:

$$\mathbf{F}_{\text{HLLC}} = \begin{cases} \mathbf{F}_L, & S_L \geq 0 \\ \mathbf{F}_L^*, & S_L \leq 0 \leq S_* \\ \mathbf{F}_R^*, & S_* \leq 0 \leq S_R \\ \mathbf{F}_R, & S_R \leq 0 \end{cases} \tag{10}$$

where S_L and S_R are estimates of the fastest left- and right-moving waves, and S_* represents the contact wave speed. The wave speeds are estimated as:

$$S_L = \min(u_L - a_L,\, u_R - a_R) \tag{11}$$

$$S_R = \max(u_L + a_L,\, u_R + a_R) \tag{12}$$

2.4 Higher-Order Reconstruction

To achieve second-order spatial accuracy while maintaining stability near discontinuities, the Monotonic Upstream-centered Scheme for Conservation Laws (MUSCL) reconstruction is combined with the HLLC Riemann solver. [6] Instead of assuming piecewise constant states, the left and right states at each interface are reconstructed using limited gradients:

$$\mathbf{U}^L_{i+\frac{1}{2}} = \mathbf{U}_i + \frac{1}{2}\phi(r_i)\,(\mathbf{U}_i - \mathbf{U}_{i-1}) \tag{13}$$

$$\mathbf{U}^R_{i+\frac{1}{2}} = \mathbf{U}_{i+1} - \frac{1}{2}\phi(r_{i+1})\,(\mathbf{U}_{i+1} - \mathbf{U}_i) \tag{14}$$

where $\phi(r)$ is a Total Variation Diminishing (TVD) slope limiter used to prevent non-physical oscillations (Gibbs phenomenon) near discontinuities. Two limiters are considered:

- **Van Leer limiter:**

$$\phi(r) = \frac{r + |r|}{1 + |r|} \tag{15}$$

  This limiter provides smooth transitions and excellent accuracy in regions of continuous flow.

- **MinMod limiter:**

$$\text{MinMod}(\Delta^-, \Delta^+) = \frac{1}{2}\left[\text{sign}(\Delta^-) + \text{sign}(\Delta^+)\right] \times \min(|\Delta^-|, |\Delta^+|) \tag{16}$$

  This limiter is slightly more diffusive but ensures strict TVD stability, making it highly robust for flows with strong terminal normal shocks.

2.5 Time Integration

To advance the semi-discrete equations in time, a Strong Stability Preserving second-order Runge–Kutta (SSP-RK2) scheme is utilized. [7] This predictor-corrector method ensures temporal stability when coupled with the spatially bounded TVD limiters. The integration proceeds as follows:

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t\, \mathbf{L}(\mathbf{U}^n) \tag{17}$$

$$\mathbf{U}^{n+1} = \frac{1}{2}\mathbf{U}^n + \frac{1}{2}\mathbf{U}^{(1)} + \frac{1}{2}\Delta t\, \mathbf{L}(\mathbf{U}^{(1)}) \tag{18}$$

where $\mathbf{L}(\mathbf{U})$ represents the spatial residual (fluxes and source terms). The time step Δt is determined dynamically at each iteration using a standard Courant–Friedrichs–Lewy (CFL) condition to satisfy numerical stability criteria.

3 Computational Setup

3.1 Nozzle Geometries

To evaluate the robustness of the numerical solver, two distinct nozzle contours are investigated. The first is a piecewise linear converging-diverging (double-cone) nozzle, which features a distinct discontinuity in the area gradient at the throat. The second is the historic Anderson parabolic nozzle, defined by a smooth, C^1-continuous quadratic profile:

$$\frac{A(x)}{A^*} = 1 + 2.2\left(\frac{x}{L} - 1.5\right)^2 \tag{19}$$

Comparing these geometries enables an assessment of the solver's ability to handle both smooth area variations and sharp geometric transitions.

3.2 Initial and Boundary Conditions

The flow is initialized with a stagnant condition (near-zero velocity) throughout the domain. The inlet thermodynamic conditions are prescribed to mimic a heated micro cold-gas thruster chamber:

$$p_{in} = 3 \times 10^5 \text{ Pa (3 bar)}, \quad T_{in} = 600 \text{ K} \tag{20}$$

At the outlet, characteristic non-reflecting boundary conditions are utilized for subsonic exits, while extrapolation is used for fully supersonic exits. The back pressure (p_{back}) is varied systematically to capture the over-expanded (shock-containing) and perfectly expanded (deep space vacuum, $p_{back} \approx$ 100 Pa) regimes.

4 Code Verification & Validation (V&V)

4.1 Grid Independence Study

A grid independence study is performed to ensure that the numerical solution is not significantly affected by the choice of spatial discretization.

The solver is run for multiple grid resolutions, and the variation of solution accuracy (quantified using Mean Absolute Percentage Error, MAPE) with the number of grid points is analyzed.

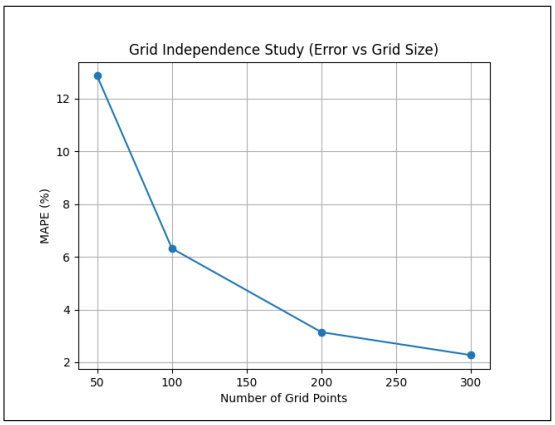

Figure 1: Grid independence study showing reduction in error (MAPE) with increasing grid resolution.

As the number of grid points increases, the error decreases significantly, indicating improved accuracy of the numerical solution. The rate of error reduction diminishes for finer grids, demonstrating convergence of the solution.

Beyond a certain grid resolution, further refinement results in only marginal improvements, indicating that the solution has become effectively grid-independent. Based on this study, an optimal grid size is selected that provides a balance between computational efficiency and solution accuracy.

4.2 Analytical Validation

To validate the numerical solver, an analytical solution for quasi-one-dimensional compressible flow through a converging–diverging nozzle is constructed using isentropic flow relations along with normal shock theory. [3]

For isentropic regions, the area–Mach number relation is given implicitly by:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{21}$$

This nonlinear equation is solved numerically at each spatial location to obtain the Mach number distribution corresponding to the given area ratio. The appropriate branch (subsonic or supersonic) is selected based on the imposed boundary conditions and expected flow regime.

For cases involving shocks, normal shock relations are used to determine the post-shock flow properties. The shock location is obtained iteratively by ensuring that the downstream pressure matches the imposed back pressure, thereby enforcing consistency between the isentropic solution and the shock jump conditions. Once the Mach number distribution is determined, other flow variables such as static pressure, density, and temperature are computed using standard isentropic relations.

Validation Metrics:

The numerical results are quantitatively compared with the analytical solution using the following error measures:

$$L_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(M_{\text{num},i} - M_{\text{exact},i}\right)^2} \tag{22}$$

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{M_{\text{num},i} - M_{\text{exact},i}}{M_{\text{exact},i}}\right| \times 100 \tag{23}$$

The MUSCL–HLLC scheme is validated against the analytical solution for all three back pressure cases. The comparison includes Mach number and static pressure distributions.

The numerical solutions show strong agreement with the analytical predictions across all flow regimes. The MUSCL–HLLC scheme provides the closest match, accurately capturing both smooth variations and sharp discontinuities.

Small deviations are observed near shock locations, which can be attributed to numerical diffusion and grid discretization effects inherent in finite-volume methods. First-order schemes, such as Lax–Friedrichs, exhibit greater smearing of discontinuities, while higher-order schemes significantly improve resolution.

Error Analysis:

The error levels are consistently lower than those observed in first-order schemes. The L_2 norm and MAPE indicate high overall accuracy, while the maximum error remains localized near the shock due to its discontinuous nature.

The use of higher-order reconstruction significantly reduces numerical diffusion, leading to improved agreement with the analytical solution in both smooth and discontinuous regions. The Min-Mod limiter used in the high back pressure case ensures stability by suppressing oscillations, albeit with a slight increase in numerical dissipation.

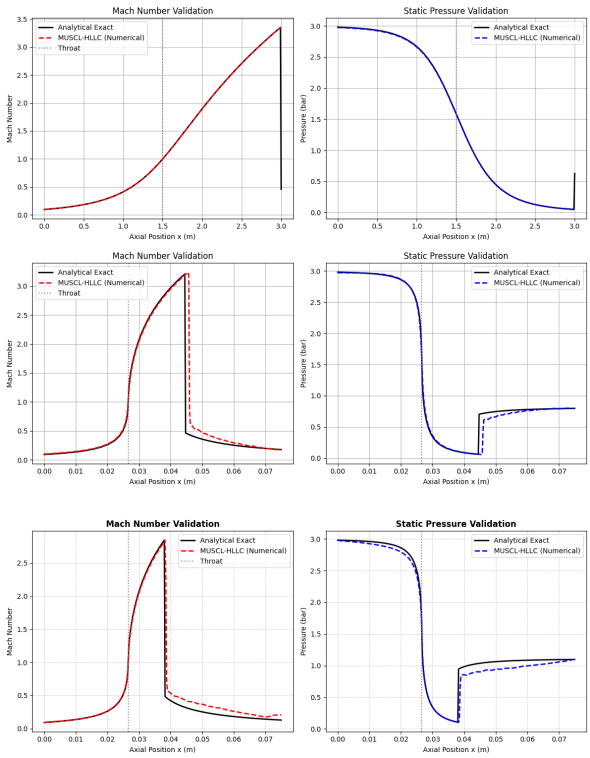

Figure 2: Comparison of numerical and analytical Mach number and pressure distributions using MUSCL–HLLC scheme for high (top), intermediate (middle), and low (bottom) back pressure cases.

```
Analytical Shock Location: x = 3.000 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 1.6410e-01
Mean Absolute Percentage Error (MAPE): 2.24%
```

```
Analytical Shock Location: x = 0.045 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 3.6028e-01
Mean Absolute Percentage Error (MAPE): 19.32%
```

```
Analytical Shock Location: x = 0.038 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 1.8243e-01
Mean Absolute Percentage Error (MAPE): 22.59%
```

Figure 3: Error metrics for the MUSCL–HLLC scheme across different back pressure cases: high (top), intermediate (middle), and low (bottom).

Overall, the validation confirms the accuracy and robustness of the implemented numerical framework for quasi-one-dimensional compressible flow simulations.

4.3 Commercial CFD Verification (ANSYS Fluent)

To further validate the numerical solver, the results are compared with simulations performed using *ANSYS Fluent*. The ANSYS data corresponds to an axisymmetric two-dimensional simulation of the same nozzle geometry under identical inlet and back pressure conditions. [8]

The ANSYS results are interpolated onto the computational grid of the one-dimensional solver to enable direct comparison. Duplicate spatial points are averaged to ensure consistency in the dataset.

Comparison of Results:

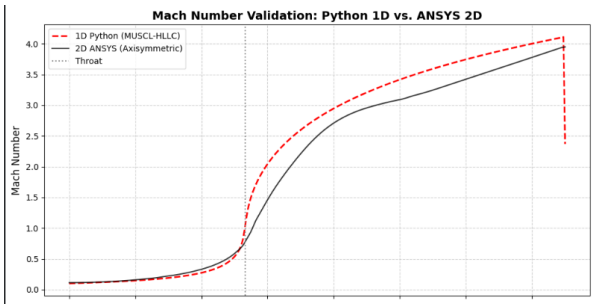

Figure 4: Comparison of Mach number distribution between 1D numerical solver (MUSCL–HLLC) and 2D axisymmetric ANSYS simulation.

The comparison shows good overall agreement between the numerical and ANSYS solutions. The Mach number distribution follows a similar trend, with both approaches capturing acceleration near the throat and supersonic expansion in the diverging section.

ANSYS Flow Field Visualization:

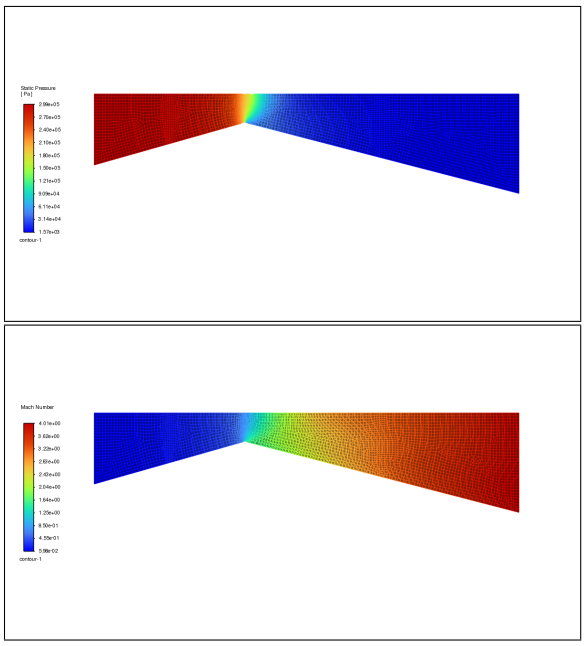

Figure 5: ANSYS Fluent contours for static pressure (top) and Mach number (bottom). The results show strong pressure gradients near the throat and supersonic expansion in the diverging section.

The contour plots provide a detailed two-

dimensional visualization of the flow field. A strong pressure gradient is observed near the throat, followed by rapid expansion in the diverging section. The Mach number contours clearly show acceleration from subsonic to supersonic regimes.

Error Analysis:

The percentage error between numerical and ANSYS results is computed as:

$$\text{Error}(\%) = \left| \frac{M_{\text{num}} - M_{\text{ANSYS}}}{M_{\text{ANSYS}}} \right| \times 100 \tag{24}$$

The computed error metrics are:

- Mean Absolute Percentage Error (MAPE): 12.60%

The deviations are most prominent near regions with strong gradients, particularly close to the throat and in the expansion region. These differences arise primarily due to the inherent limitations of the quasi-one-dimensional model, which does not capture multidimensional effects such as radial gradients and boundary layer development present in the axisymmetric ANSYS simulation.

Despite these differences, the solver accurately captures the dominant flow physics, including acceleration through the throat and supersonic expansion. This demonstrates that the developed quasi-one-dimensional model provides a reliable approximation for nozzle flow analysis while maintaining significantly lower computational cost compared to full CFD simulations.

5 Results and Flow Physics Discussion

The numerical simulations are performed for three different back pressure conditions, corresponding to distinct flow regimes: fully subsonic flow, shock-containing flow, and fully supersonic flow. The results are analyzed in terms of Mach number and density distributions, along with the ability of different numerical schemes to capture key flow features.

5.1 Comparison of Numerical Schemes

All three numerical methods successfully capture the overall flow behavior across different regimes. However, significant differences are observed in their ability to resolve discontinuities.

The Lax–Friedrichs scheme, due to its inherently diffusive nature, produces smeared shock profiles and smoothened gradients. While it remains stable and robust, it lacks the ability to sharply resolve discontinuities.

The HLLC scheme significantly improves shock resolution by accurately capturing wave structures, including the contact discontinuity. Shocks are represented more sharply compared to the Lax–Friedrichs method, with reduced numerical diffusion.

The MUSCL–HLLC scheme further enhances the solution by providing second-order spatial accuracy. It captures shocks with high sharpness while maintaining smooth and accurate profiles in expansion regions. The use of flux limiters prevents non-physical oscillations near discontinuities, ensuring both accuracy and stability.

In some cases, minor oscillations are observed near shock regions in the HLLC scheme, which are effectively controlled in the MUSCL formulation through slope limiting.

5.1.1 Lax–Friedrichs Scheme

The Lax–Friedrichs scheme is first evaluated for three different back pressure conditions to study its ability to capture various flow regimes.

High Back Pressure ($p_{back} = 1.1 \times 10^5$ Pa):

For the highest back pressure case, the flow initially accelerates from subsonic to supersonic conditions near the throat, as indicated in Figure 7. The Mach number crosses unity at the throat and reaches a supersonic peak (around M2), after which a normal shock forms in the diverging section. Downstream of the shock, the flow remains subsonic and gradually decelerates toward the exit. The density variation (Figure 6) reflects this behavior: density decreases as the flow accelerates toward the throat, reaches a minimum in the supersonic region, and then exhibits a sharp rise across the shock, followed by a smoother variation in the downstream subsonic region.

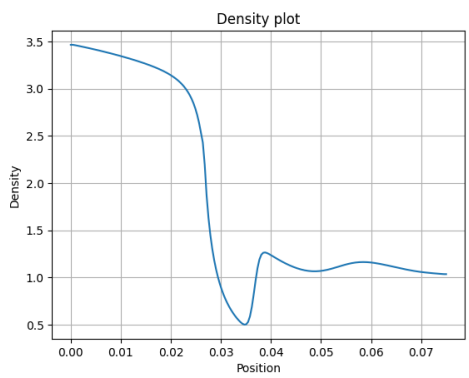

Figure 6: Density distribution for high back pressure using Lax–Friedrichs scheme (subsonic flow).

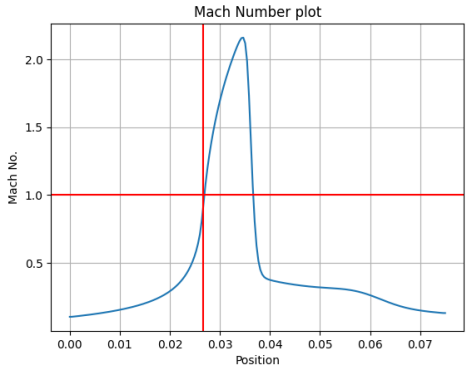

Figure 7: Mach number distribution for high back pressure using Lax–Friedrichs scheme.

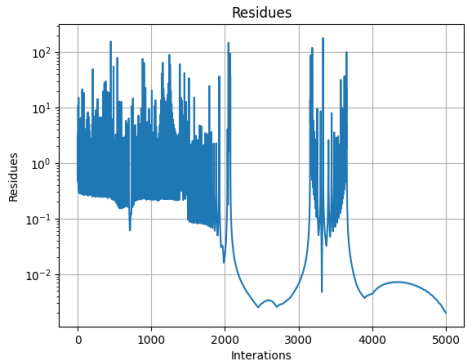

Figure 8: Residual convergence history for high back pressure case.

Intermediate Back Pressure ($p_{back} = 5 \times 10^4$ Pa):

For intermediate back pressure, the flow accelerates to supersonic conditions near the throat, followed by the formation of a normal shock in the diverging section. This is observed in Figure 10 as a gradual transition due to numerical diffusion. The density plot (Figure 9) shows a smeared rise across the shock region.

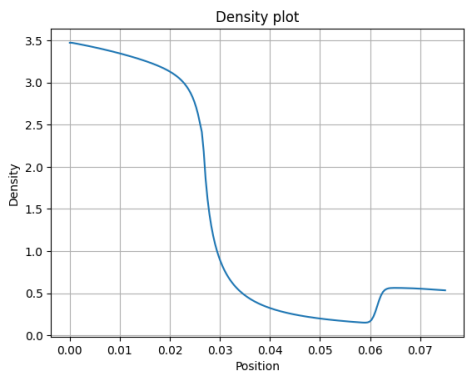

Figure 9: Density distribution for intermediate back pressure (shock case) using Lax–Friedrichs scheme.

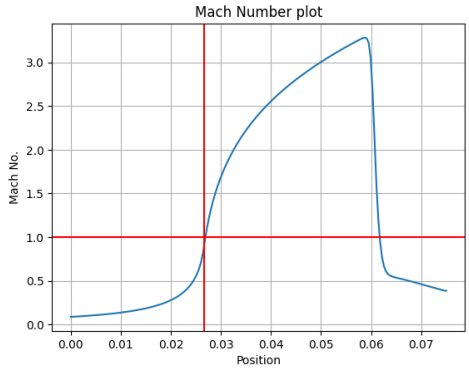

Figure 10: Mach number distribution for intermediate back pressure showing smeared shock.

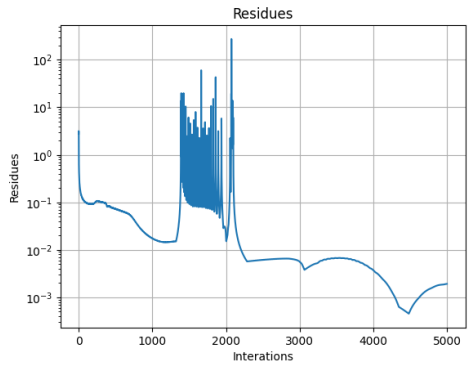

Figure 11: Residual convergence history for intermediate back pressure case.

Low Back Pressure ($p_{back} = 5 \times 10^3$ Pa): For the lowest back pressure, the flow becomes fully supersonic downstream of the throat. The Mach number (Figure 13) increases monotonically along the diverging section, while the density (Figure 12) decreases continuously, indicating expansion without shock formation.

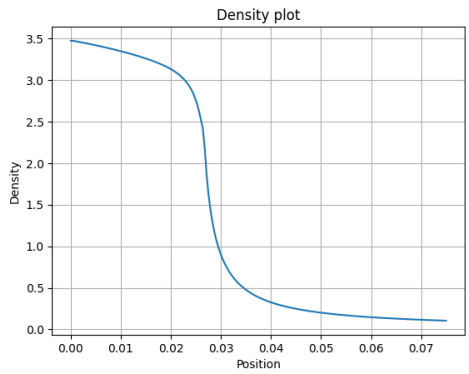

Figure 12: Density distribution for low back pressure (supersonic flow) using Lax–Friedrichs scheme.

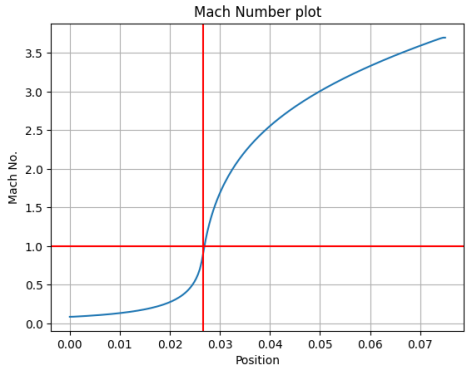

Figure 13: Mach number distribution for low back pressure showing supersonic expansion.

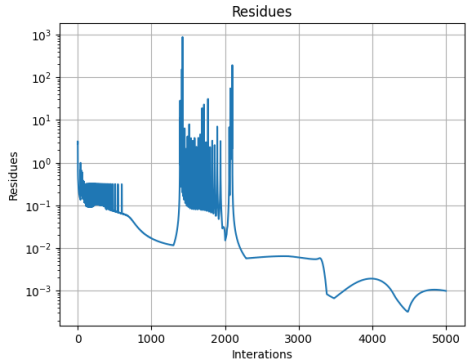

Figure 14: Residual convergence history for low back pressure case.

Quadratic Nozzle (Intermediate Back Pressure):

The Lax–Friedrichs scheme is also tested on a smooth quadratic nozzle geometry for the intermediate back pressure case. The results show similar overall trends; however, the solution appears smoother due to the continuous area variation. Unlike the linear nozzle, no artificial kink is observed near the throat.

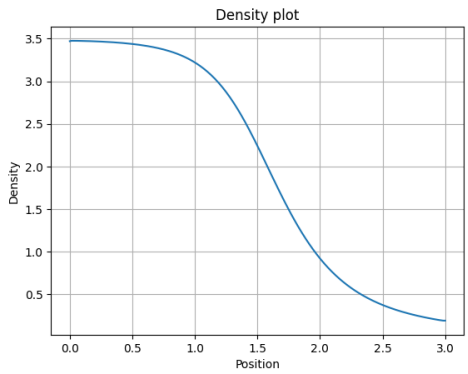

Figure 15: Density distribution for quadratic nozzle (intermediate back pressure).

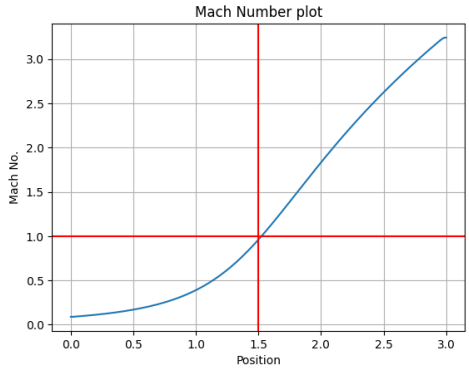

Figure 16: Mach number distribution for quadratic nozzle.

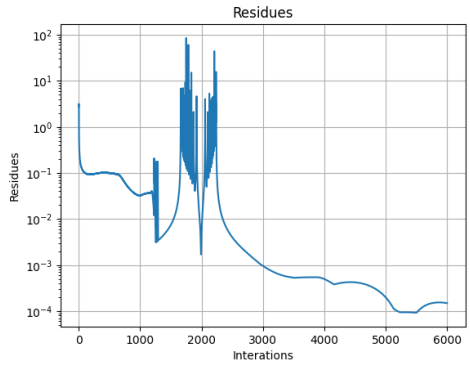

Figure 17: Residual convergence history for quadratic nozzle case.

Overall, the Lax–Friedrichs scheme successfully captures the qualitative flow behavior across all regimes. However, due to its inherently diffusive nature, sharp features such as shocks are smeared over a wider region, and gradients are significantly smoothened. This limits its accuracy in resolving discontinuities compared to higher-order methods.

5.1.2 HLLC Scheme

The HLLC (Harten–Lax–van Leer–Contact) scheme is evaluated for three different back pressure conditions to assess its capability in accurately resolving compressible flow features, particularly shocks and expansion regions.

High Back Pressure ($p_{back} = 1.5 \times 10^5$ Pa):

For the highest back pressure, the flow remains predominantly subsonic throughout the nozzle. The Mach number increases smoothly toward the throat but does not reach strongly supersonic values. The static pressure decreases monotonically along the nozzle, consistent with subsonic compressible flow behavior.

The solution remains smooth without any discontinuities, indicating that no shock is formed within the nozzle.

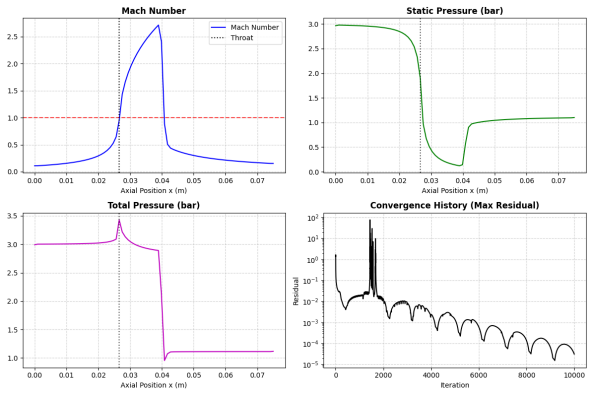

Figure 18: Flow variables for high back pressure using HLLC scheme.

Intermediate Back Pressure ($p_{back} = 8 \times 10^4$ Pa):

For intermediate back pressure, the flow accelerates to supersonic conditions near the throat and undergoes a normal shock in the diverging section. This is clearly captured in the Mach number distribution as a sharp drop from supersonic to subsonic values.

Compared to the Lax–Friedrichs scheme, the shock is significantly sharper with minimal numerical smearing. The static pressure shows a corresponding abrupt rise across the shock location, indicating correct shock capturing behavior.

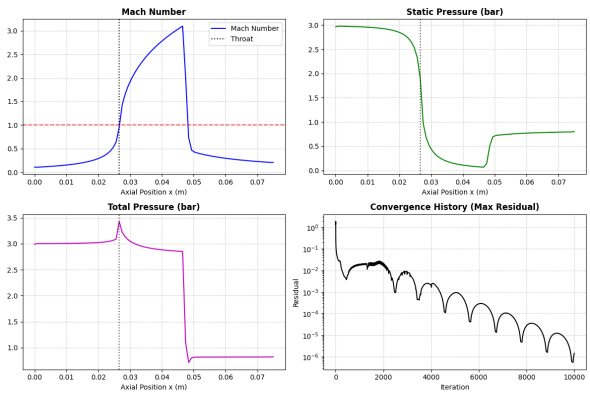

Figure 19: Flow variables for intermediate back pressure showing shock formation using HLLC scheme.

Low Back Pressure ($p_{back} = 5 \times 10^3$ Pa):

For the lowest back pressure, the flow becomes fully supersonic downstream of the throat. The Mach number increases monotonically along the diverging section, and the static pressure continuously decreases, indicating supersonic expansion without shock formation.

The HLLC scheme accurately captures the expansion behavior with minimal numerical diffusion.

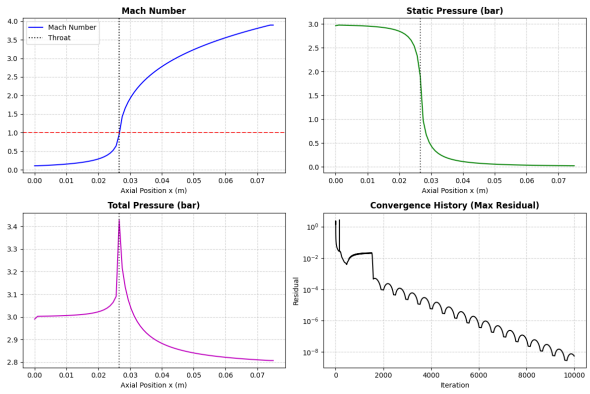

Figure 20: Flow variables for low back pressure showing fully supersonic expansion using HLLC scheme.

Analytical Validation:

The numerical results obtained using the HLLC scheme are validated against the analytical quasi-one-dimensional solution for all three back pressure cases. The comparison includes both shock-free and shock-containing regimes.

In all cases, the numerical solution closely follows the analytical profiles in both subsonic and supersonic regions. For shock-containing cases, the location and strength of the normal shock are accurately predicted. Minor discrepancies are observed near the shock due to numerical diffusion and finite grid resolution.

Error Analysis:

The accuracy of the HLLC scheme is evaluated using standard error metrics computed over all back pressure cases. The results are summarized in Figure 22.

The computed errors indicate that the HLLC scheme provides good overall agreement with the analytical solution. The relatively high maximum error is primarily localized near the shock region, where small spatial shifts in shock position lead to large pointwise deviations.

Away from the shock, the numerical solution closely matches the analytical solution, resulting in low average error. This behavior is consistent across different back pressure cases and highlights the robustness of the HLLC scheme in capturing both smooth and discontinuous flow features.

Overall, the HLLC scheme provides a significant improvement over the Lax–Friedrichs method. It captures shocks sharply, reduces numerical diffusion, and maintains good accuracy in both smooth and discontinuous regions. Minor oscillations or deviations near the shock are attributed to wave speed estimation and grid resolution effects.

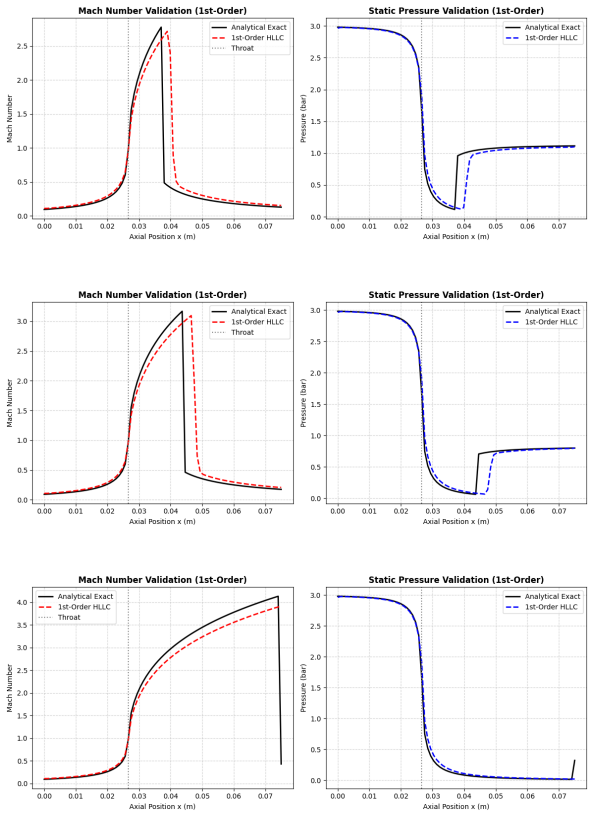

Figure 21: Comparison of numerical and analytical Mach number and pressure distributions for three back pressure cases using the HLLC scheme: high (top), intermediate (middle), and low (bottom).

```
--- 1st-Order Solver Validation Metrics ---
L2 Norm Error (Mach): 4.2492e-01
Mean Absolute Percentage Error (MAPE): 34.02%
```

```
--- 1st-Order Solver Validation Metrics ---
L2 Norm Error (Mach): 5.4612e-01
Mean Absolute Percentage Error (MAPE): 41.09%
```

```
--- 1st-Order Solver Validation Metrics ---
L2 Norm Error (Mach): 4.2037e-01
Mean Absolute Percentage Error (MAPE): 18.32%
```

Figure 22: Error metrics for the HLLC scheme across different back pressure cases: high (top), intermediate (middle), and low (bottom), showing L_2 norm, MAPE, and maximum percentage error.

5.1.3 MUSCL–HLLC Scheme

The MUSCL–HLLC scheme is evaluated under different back pressure conditions and nozzle geometries to assess its higher-order accuracy and ability to resolve flow features with reduced numerical diffusion.

Low Back Pressure ($p_{back} = 5 \times 10^3$ Pa, Quadratic Nozzle):

For the lowest back pressure, the flow becomes fully supersonic downstream of the throat. The Mach number increases smoothly and monotonically along the diverging section, while density and pressure decrease continuously.

Due to the smooth quadratic geometry and second-order spatial accuracy of the MUSCL scheme, the solution exhibits very smooth profiles without any numerical artifacts. Expansion regions are captured with high fidelity, and no oscillations are observed.

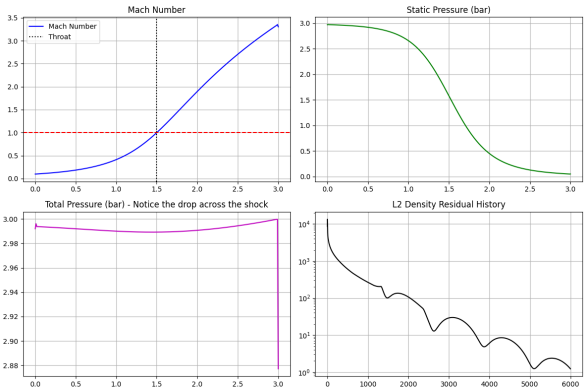

Figure 23: Flow variables for low back pressure using MUSCL–HLLC scheme (quadratic nozzle).

Intermediate Back Pressure ($p_{back} = 5 \times 10^4$ Pa, Double-Cone Nozzle):

For intermediate back pressure, the flow accelerates to supersonic conditions near the throat, followed by the formation of a normal shock in the diverging section.

The MUSCL–HLLC scheme captures the shock sharply with minimal numerical smearing compared to first-order methods. The Mach number shows a steep drop across the shock, and the pressure exhibits a corresponding abrupt rise.

The effect of nozzle geometry is also visible. Due to the piecewise linear nature of the double-cone nozzle, a slight kink may be observed near the throat, associated with the discontinuity in the area gradient.

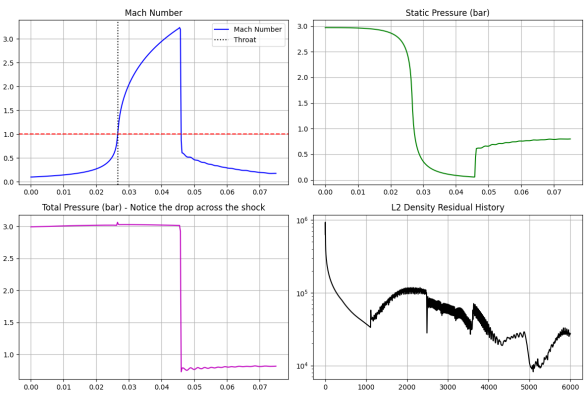

Figure 24: Flow variables for intermediate back pressure showing shock formation using MUSCL–HLLC scheme.

High Back Pressure ($p_{back} = 1.1 \times 10^5$ Pa, Double-Cone Nozzle, MinMod Limiter):

For the highest back pressure, the flow remains predominantly subsonic, with smooth variations in flow variables. However, due to the presence of steep gradients near the throat region, a MinMod limiter is employed to ensure numerical stability and prevent non-physical oscillations.

The MinMod limiter effectively suppresses spurious oscillations, although it introduces slightly higher numerical dissipation compared to smoother limiters. Despite this, the solution remains stable and physically consistent.

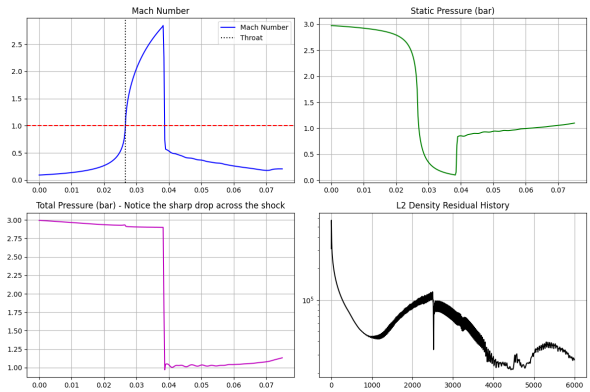

Figure 25: Flow variables for high back pressure using MUSCL–HLLC scheme with MinMod limiter.

The MUSCL–HLLC scheme is validated against the analytical solution for all three back pressure cases. The comparison includes Mach number and static pressure distributions.

Analytical Validation:

The MUSCL–HLLC results show excellent agreement with the analytical solution across all flow regimes. The shock location is captured accurately, and the transition across the shock is significantly sharper compared to first-order methods.

Error Analysis:

```
Analytical Shock Location: x = 3.000 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 1.6410e-01
Mean Absolute Percentage Error (MAPE): 2.24%
```

```
Analytical Shock Location: x = 0.045 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 3.6028e-01
Mean Absolute Percentage Error (MAPE): 19.32%
```

```
Analytical Shock Location: x = 0.038 m

--- 2nd-Order (MUSCL) Solver Validation Metrics ---
L2 Norm Error (Mach): 1.8243e-01
Mean Absolute Percentage Error (MAPE): 22.59%
```

Figure 27: Error metrics for the MUSCL–HLLC scheme across different back pressure cases: high (top), intermediate (middle), and low (bottom).

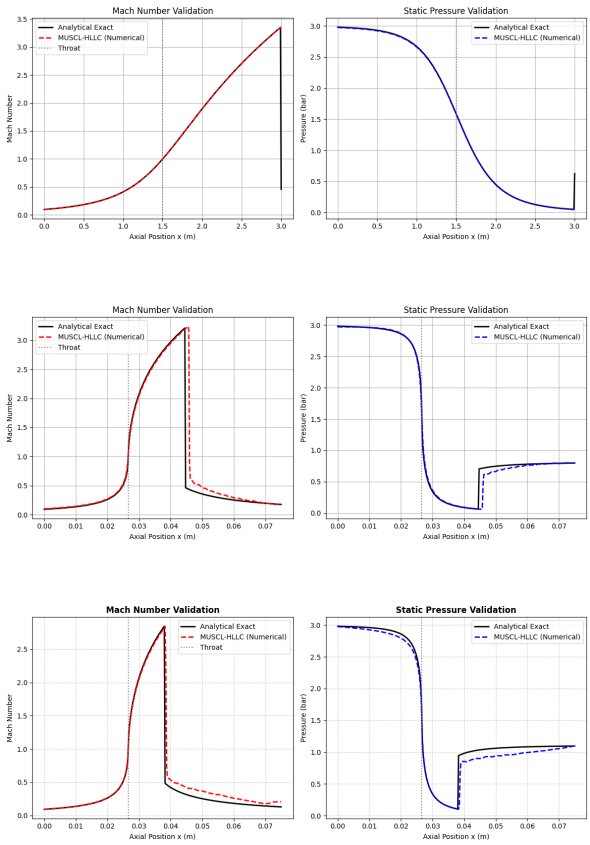

Figure 26: Comparison of numerical and analytical Mach number and pressure distributions using MUSCL–HLLC scheme for high (top), intermediate (middle), and low (bottom) back pressure cases.

The error levels are consistently lower than those observed in first-order schemes. The L_2 norm and MAPE indicate high overall accuracy, while the maximum error remains localized near the shock due to its discontinuous nature.

The use of higher-order reconstruction significantly reduces numerical diffusion, leading to improved agreement with the analytical solution in both smooth and discontinuous regions. The MinMod limiter used in the high back pressure case ensures stability by suppressing oscillations, albeit with a slight increase in numerical dissipation.

5.1.4 Discussion:

Compared to the Lax–Friedrichs and first-order HLLC schemes, the MUSCL–HLLC method provides significantly improved resolution of both smooth and discontinuous flow features. Shocks are captured sharply with minimal numerical diffusion, and expansion regions are represented with high accuracy.

The influence of nozzle geometry becomes more pronounced in this higher-order scheme. While the quadratic nozzle produces smooth and continuous

profiles, the double-cone nozzle introduces slight discontinuities in gradients near the throat, which are reflected in the solution.

The choice of flux limiter also plays an important role. The Van Leer limiter allows higher accuracy in smooth regions, while the MinMod limiter ensures robustness in cases with stronger gradients by preventing oscillations.

Overall, the MUSCL–HLLC scheme provides the most accurate and physically consistent results among all methods considered, achieving a balance between high resolution and numerical stability.

5.2 Influence of Back Pressure

For the highest back pressure, the flow remains entirely subsonic throughout the nozzle. The Mach number increases in the converging section, reaches a maximum near the throat, and decreases in the diverging section. The density correspondingly decreases toward the throat and increases downstream, consistent with subsonic compressible flow behavior.

For intermediate back pressure, the flow accelerates to supersonic conditions near the throat, followed by the formation of a normal shock in the diverging section. This is observed as a sharp drop in Mach number from supersonic to subsonic values. The density shows a corresponding discontinuous rise across the shock, indicating strong compression.

For the lowest back pressure, the flow becomes fully supersonic after the throat and remains supersonic throughout the diverging section. The Mach number increases monotonically, while the density continuously decreases along the nozzle, indicating expansion without shock formation.

The location and sharpness of the shock are strongly influenced by both back pressure and numerical scheme. As the back pressure decreases, the shock moves downstream and eventually exits the nozzle, leading to fully supersonic flow.

The MUSCL–HLLC scheme provides the most accurate prediction of shock location and strength, closely matching expected physical behavior. The Lax–Friedrichs scheme tends to diffuse the shock over a larger region, while the HLLC scheme offers a good balance between accuracy and computational robustness.

Overall, the results demonstrate that while all methods capture the qualitative flow features, higher-order schemes significantly improve quantitative accuracy, shock resolution, and representation of smooth flow regions. The combination of MUSCL reconstruction with the HLLC solver provides the best performance among the methods considered.

5.3 Influence of Nozzle Contour

The influence of nozzle geometry is examined by comparing results obtained using a piecewise linear (double-cone) nozzle and a smooth quadratic nozzle.

For the Lax–Friedrichs scheme, the effect of geometry is relatively less pronounced due to numerical diffusion, which smoothens sharp geometric variations.

In contrast, higher-order schemes such as HLLC and MUSCL–HLLC clearly capture the influence of geometry. In the case of the linear nozzle, a slight kink is observed in the density distribution near the throat, arising from the discontinuity in the area gradient. The quadratic nozzle, being smooth and differentiable, results in smooth variations in all flow variables without such artifacts.

6 Engineering Application: Thruster Performance

6.1 Physical Scaling

To demonstrate the practical applicability of the developed numerical solver, the dimensionless quasi-one-dimensional flow solution is scaled to represent a realistic micro-thruster configuration.

A physical throat area of 1 mm^2 is assumed, and the exit flow properties obtained from the numerical simulations are used to evaluate key performance parameters.

The mass flow rate is computed as:

$$\dot{m} = \rho_{exit}\, u_{exit}\, A_{exit} \tag{25}$$

The thrust generated by the nozzle consists of both momentum and pressure contributions:

$$F = \dot{m}\, u_{exit} + (p_{exit} - p_{back}) A_{exit} \tag{26}$$

The specific impulse, which represents the efficiency of the propulsion system, is evaluated as:

$$I_{sp} = \frac{F}{\dot{m}\, g_0} \tag{27}$$

6.2 Propellant Parametric Study

The computed performance metrics for different working fluids and operating conditions are summarized below:

Table 1: Thruster Performance Metrics (Quadratic Nozzle)

Case	$\dot{m}$ (g/s)	Thrust (mN)	I_{sp} (s)
Nitrogen	0.48	447.51	94.40
Helium	0.20	451.14	234.76

Table 2: Thruster Performance Metrics (Double-Cone Nozzle)

Case	$\dot{m}$ (g/s)	Thrust (mN)	I_{sp} (s)
Helium	0.20	481.90	244.06
Nitrogen	0.50	495.57	100.71

The results clearly highlight the influence of working fluid properties on thruster performance. Helium, due to its lower molecular weight and higher speed of sound, achieves significantly higher exhaust velocities, resulting in substantially higher specific impulse compared to nitrogen.

In contrast, nitrogen exhibits higher mass flow rates due to its greater density, leading to comparable thrust levels despite lower exhaust velocities. This reflects the classical trade-off between thrust and efficiency in cold-gas propulsion systems.

The predicted thrust levels (approximately 450–500 mN) and specific impulse values (up to ~ 244 s) are consistent with typical performance ranges reported for micro cold-gas thrusters used in small satellite and CubeSat application [1, 9].

It should be noted that the present analysis assumes ideal, inviscid flow conditions. In practical systems, additional losses due to viscous effects, boundary layer growth, heat transfer, and non-ideal gas behavior may reduce the actual performance. Nevertheless, the present results provide a reliable first-order estimate of thruster characteristics and demonstrate the capability of the numerical framework for propulsion analysis.

7 Conclusion

In this work, a quasi-one-dimensional numerical solver for compressible flow through converging–diverging nozzles has been developed and systematically evaluated using multiple finite volume schemes, including Lax–Friedrichs, HLLC, and MUSCL–HLLC methods.

The study demonstrates that while the Lax–Friedrichs scheme provides robust and stable solutions, it suffers from excessive numerical dissipation, leading to smeared shock profiles and reduced accuracy. The HLLC scheme significantly improves shock resolution by accurately capturing wave structures, offering a better balance between stability and accuracy. The MUSCL–HLLC scheme further enhances solution quality by incorporating second-order spatial reconstruction, resulting in sharp shock capturing, reduced numerical diffusion, and improved resolution in smooth regions.

A comprehensive parametric study was performed by varying back pressure conditions, successfully capturing subsonic, shock-containing, and fully supersonic flow regimes. The influence of nozzle geometry was also investigated, where the smooth quadratic nozzle produced physically consistent and smooth solutions, while the piecewise linear nozzle introduced minor artifacts near the throat due to discontinuities in area gradient.

Validation against analytical solutions confirmed the accuracy of the numerical methods, with the MUSCL–HLLC scheme showing the closest agreement. Additional comparison with ANSYS Fluent simulations demonstrated that the quasi-one-dimensional solver effectively captures the dominant flow physics, with deviations primarily arising from multidimensional effects not accounted for in the simplified model.

The grid independence study confirmed convergence of the numerical solution, ensuring reliability of the results beyond a certain resolution. Furthermore, physical scaling of the solution to a micro-thruster configuration yielded realistic thrust and specific impulse values, demonstrating the practical applicability of the developed solver for propulsion system analysis.

Overall, the MUSCL–HLLC scheme is identified as the most accurate and efficient method among those implemented, providing high-fidelity results with relatively low computational cost. The developed framework serves as a reliable tool for preliminary nozzle design and performance evaluation in aerospace applications, particularly for small satellite propulsion systems.

References

[1] Lokesh Meena, M. S. Niranjan, Aman, Gautam, Gagandeep, Gaurav Kumar, and M. Zunaid. Numerical study of convergent-divergent nozzle at different throat diameters and divergence angles. *Materials Today: Proceedings*, 46:10676–10680, 2021.

[2] Jesse Chan, Khemraj Shukla, Xinhui Wu, Ruofeng Liu, and Prani Nalluri. High order entropy stable schemes for the quasi-one-dimensional shallow water and compressible euler equations. *arXiv preprint arXiv:2307.12089*, 2024.

[3] John D. Anderson. *Modern Compressible Flow: With Historical Perspective.* McGraw-Hill, New York, 2nd edition, 1990.

[4] R. Eymard, T. Gallouët, R. Herbin, and J.-C. Latché. Finite volume schemes and lax–wendroff consistency. *ESAIM: Mathematical Modelling and Numerical Analysis*, 2021.

[5] Eleuterio F Toro. *Riemann solvers and numerical methods for fluid dynamics: a practical introduction.* Springer Science & Business Media, 2013.

[6] Bram Van Leer. Towards the ultimate conservative difference scheme. ii. monotonicity and conservation combined in a second-order scheme. *Journal of computational physics*, 14(4):361–370, 1974.

[7] Antony Jameson, Wolfgang Schmidt, and Eli Turkel. Numerical solution of the euler equations by finite volume methods using runge-kutta time-stepping schemes. In *AIAA 14th Fluid and Plasma Dynamics Conference*. American Institute of Aeronautics and Astronautics, 1981. AIAA Paper 81-1259.

[8] P. K. Tucker and S. A. Warsi. Axisymmetric computational fluid dynamics analysis of a film/dump-cooled rocket nozzle plume. In *29th Joint Propulsion Conference and Exhibit*, Monterey, California, 1993. American Institute of Aeronautics and Astronautics (AIAA). AIAA Paper 93-2349.

[9] Cahya Amalinadhi Putra. Numerical solutions of quasi-one-dimensional convergent-divergent nozzle flows. Technical report, Bandung Institute of Technology, 2020. Quasi-1D nozzle flow solved using MacCormack scheme.

Work Distribution

- **Sidhwa Tanuj Haresh (Lead Numerical Developer):** Led the development of an in-house quasi-one-dimensional finite volume solver in Python, architecting the core computational framework. Implemented the HLLC approximate Riemann solver and integrated second-order MUSCL reconstruction with TVD limiters to enhance solution accuracy and stability. Additionally, prepared and formatted the accompanying LaTeX manuscript for technical documentation.

- **Tejas Santosh Bhavekar (Commercial CFD Validation):** Led the commercial software verification phase by setting up a 2D axisymmetric computational domain in ANSYS Fluent. Generated and refined the computational mesh, configured the density-based solver under consistent boundary conditions, and extracted centerline Mach number and pressure distributions for rigorous benchmark comparison against in-house results.

- **Siddharth Vezzu (Analytical Formulation & Error Analysis):** Directed the analytical validation of the project by formulating the implicit isentropic area–Mach relations and normal shock jump conditions. Developed the exact solution baseline through custom code implementation and evaluated model accuracy by computing (L_2) norm and MAPE error metrics across multiple grid resolutions.

- **Sudarsanam Sai Sri Narasimha (Physical Scaling & Parametric Study):** Managed the aerospace application phase by defining boundary conditions for a micro-thruster model and researching the thermodynamic properties of nitrogen and helium propellants. Scaled dimensionless simulation outputs to physical dimensions (1 mm^2 throat area) and computed key performance metrics, including specific impulse and thrust.